

Chapter 7 Orthogonality and Orthogonal Projections

7.1 Introduction

One of the most important ideas in Linear Algebra is **orthogonality**. It provides the mathematical foundation for many techniques used in Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, Signal Processing, and Data Compression.

Orthogonality helps us answer questions such as:

- How similar are two vectors?
- How can redundant information be removed?
- How can high-dimensional data be projected into lower dimensions?
- How does Principal Component Analysis (PCA) reduce dimensionality?
- Why are embeddings compared using cosine similarity?
- How does the Transformer attention mechanism compare word embeddings?

Many AI algorithms rely on the fact that orthogonal vectors carry **independent information**.

7.2 Orthogonality

Definition

Two vectors are said to be **orthogonal** if they are **perpendicular** to each other.

Mathematically,

$$u \cdot v = 0$$

where

- u = first vector
- v = second vector
- $\cdot$ denotes the **dot product**

If the dot product equals zero, the vectors are orthogonal.

Example 1

Consider

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The dot product is

$$u \cdot v = (1)(0) + (0)(1) = 0$$

Therefore,

$$u \perp v$$

These vectors are orthogonal.

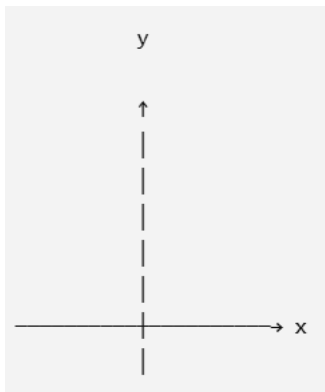
Geometric Interpretation

Imagine standing at the origin.

One vector points **east**.

The other points **north**.

They form a **90° angle**.



These directions are completely independent.

7.3 Dot Product and Angle Between Vectors

The dot product can also be written as

$$u \cdot v = ||u|| ||v|| \cos \theta$$

where

- $||u||$ is the magnitude of vector u
- $||v||$ is the magnitude of vector v
- θ is the angle between the vectors

Important Cases

If

$$\theta = 0^\circ$$

then

$$\cos 0 = 1$$

Maximum similarity.

If

$$\theta = 90^\circ$$

then

$$\cos 90 = 0$$

Orthogonal vectors.

If

$$\theta = 180^\circ$$

then

$$\cos 180 = -1$$

Vectors point in opposite directions.

7.4 Why Orthogonality Matters

Suppose we measure two quantities:

- Student Height
- Student Weight

These variables are somewhat related.

Now consider

- Height

- Blood Group

These variables are essentially independent.

Independent variables are closer to being **orthogonal**.

Orthogonality therefore represents **independent information**.

In Artificial Intelligence

Orthogonality is desirable because it reduces redundancy. Instead of learning the same information repeatedly,

AI models prefer features that capture different aspects of the data.

Examples include:

- Principal Component Analysis (PCA)
 - Independent Component Analysis (ICA)
 - QR Decomposition
 - Singular Value Decomposition (SVD)
 - Transformer embeddings
 - Recommendation systems
-

7.5 Orthonormal Vectors

Orthogonal vectors are perpendicular.

Orthonormal vectors are

1. Orthogonal
2. Unit length

Mathematically,

$$\|v\| = 1$$

Example

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Each vector has

$$\|u\| = 1$$

and

$$\|v\| = 1$$

Hence they are **orthonormal**.

Why Unit Length?

If vectors have different magnitudes,
the larger vector dominates calculations.

Normalizing vectors removes this bias.

Example

Suppose

Student A

Income = ₹50,000

Experience = 2 years

Student B

Income = ₹500,000

Experience = 3 years

Without normalization,

Income dominates.

After normalization,

both features contribute equally.

7.6 Normalization

Given

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Magnitude

$$\|x\| = \sqrt{3^2 + 4^2} = 5$$

Normalized vector

$$\hat{x} = \frac{x}{\|x\|}$$

Therefore,

$$\hat{x} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

Now

$$\|\hat{x}\| = 1$$

7.7 Cosine Similarity

One of the most widely used similarity measures in AI is **cosine similarity**.

It measures the cosine of the angle between two vectors.

The cosine similarity between vectors u and v is

$$\text{Cosine Similarity} = \frac{u \cdot v}{\|u\| \|v\|}$$

Interpretation

Cosine Similarity Meaning

+1	Identical direction
0	Orthogonal (independent)
-1	Opposite direction

Why Cosine Similarity is Important

Modern AI systems compare vectors rather than raw text.

For example:

"Dog"



Embedding

↓

[0.12, 0.55, -0.23, ...]

Similarly,

"Puppy"

↓

Embedding

↓

[0.15, 0.53, -0.20, ...]

The vectors for **Dog** and **Puppy** point in similar directions, resulting in a cosine similarity close to **1**.

In contrast, the vectors for **Dog** and **Car** are much less aligned, giving a lower cosine similarity.

Cosine similarity is widely used in:

- Semantic Search
- Retrieval-Augmented Generation (RAG)
- Recommendation Systems
- Large Language Models
- Sentence Embeddings
- Document Similarity
- Vector Databases